

Ex: 10 WIRESHARK

AIM:

To capture, analyze, and interpret network traffic using Wireshark for the purpose of troubleshooting, security auditing, or protocol learning.

STEPS:

1. Install and Launch Wireshark

- Download from [wireshark.org](https://www.wireshark.org)
- Install and run with administrative privileges

2. Start Packet Capture

- Select the appropriate network interface (e.g., Wi-Fi, Ethernet)
- Click the blue shark fin icon to begin capturing live traffic

3. Apply Display Filters

- Use filters to isolate specific traffic:
 - `http` → HTTP traffic
 - `ip.addr == 192.0.2.1` → Specific IP
 - `tcp.port == 53` → HTTPS traffic

4. Inspect Packets

- Click on a packet to view:
 - Frame details (Ethernet, IP, TCP/UDP)
 - Application layer data (e.g., HTTP headers, DNS queries)

5. Follow Streams

- Right-click a packet → “Follow → TCP Stream” to reconstruct conversations

6. Save and Load Captures

- Save: File → Save As (`.pcapng`)
- Load: File → Open to analyze later

7. Use Color Coding and Statistics

- View → Coloring Rules to highlight traffic types
- Analyze → Protocol Hierarchy or Conversations for traffic summaries

8. Troubleshoot or Audit

- Look for anomalies: retransmissions, malformed packets, suspicious IPs
- Use filters and stream following to trace issues or threats

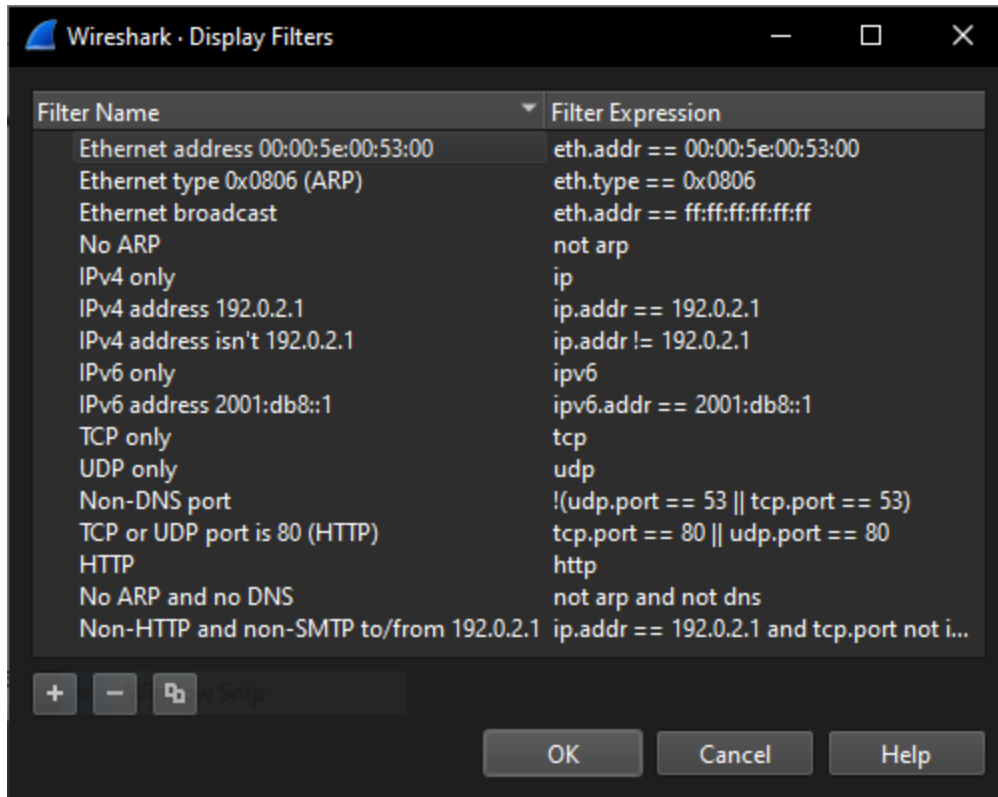
OUTPUT:

The screenshot displays the Wireshark network protocol analyzer interface. The main pane shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The packets are color-coded: green for standard traffic, yellow for ARP, and red for TCP ZeroWindow events. Packet 7167 is highlighted in red, indicating a TCP ZeroWindow event. The bottom pane shows the details of the selected packet (Frame 1), including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol fields. The status bar at the bottom indicates that 7178 packets were captured and 0 were dropped.

No.	Time	Source	Destination	Protocol	Length	Info
7159	14.352158	172.16.10.98	49.44.131.40	TCP	54	50496 → 443 [ACK] S
7160	14.352167	49.44.131.40	172.16.10.98	TCP	1514	443 → 50496 [ACK] S
7161	14.352182	49.44.131.40	172.16.10.98	TCP	1514	443 → 50496 [ACK] S
7162	14.352187	172.16.10.98	49.44.131.40	TCP	54	50496 → 443 [ACK] S
7163	14.352191	49.44.131.40	172.16.10.98	TCP	1514	443 → 50496 [ACK] S
7164	14.352246	49.44.131.40	172.16.10.98	TCP	1514	443 → 50496 [ACK] S
7165	14.352246	49.44.131.40	172.16.10.98	TCP	1514	443 → 50496 [ACK] S
7166	14.352246	49.44.131.40	172.16.10.98	TCP	106	443 → 50496 [PSH, A
7167	14.352255	172.16.10.98	49.44.131.40	TCP	54	[TCP ZeroWindow] 50
7168	14.421678	HP_35:10:d0	Broadcast	ARP	60	Who has 172.16.8.18
7169	14.437628	Dell_35:11:1f	Broadcast	ARP	60	Who has 172.16.11.1
7170	14.460392	172.16.10.52	224.0.0.251	MDNS	246	Standard query resp
7171	14.460544	fe80::8f01:f6a2:bc7...	ff02::fb	MDNS	266	Standard query resp
7172	14.460965	172.16.10.52	224.0.0.251	MDNS	83	Standard query 0x00
7173	14.461096	fe80::8f01:f6a2:bc7...	ff02::fb	MDNS	103	Standard query 0x00
7174	14.476194	172.16.9.19	172.16.11.255	NBNS	92	Name query NB DESK
7175	14.530699	Dell_34:d4:be	Broadcast	ARP	60	Who has 172.16.8.46
7176	14.563951	49.44.131.40	172.16.10.98	TCP	60	[TCP Keep-Alive] 44
7177	14.563971	172.16.10.98	49.44.131.40	TCP	54	[TCP ZeroWindow] 50
7178	14.636556	Netgear 5e:9a:f2	Nearest-Customer-Br...	STP	60	RST, Root = 32768/0

Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0
Ethernet II, Src: MicroStarINT_c5:cb:78 (d8:bb:c1:c5:cb:78), Dst: Sop...
Internet Protocol Version 4, Src: 172.16.10.98, Dst: 49.44.131.40
Transmission Control Protocol, Src Port: 50496, Dst Port: 443, Seq: ...

wireshark_Ethernet0J7EE3.pcapng Packets: 7178 - Dropped: 0 (0.0%) Profile: Default



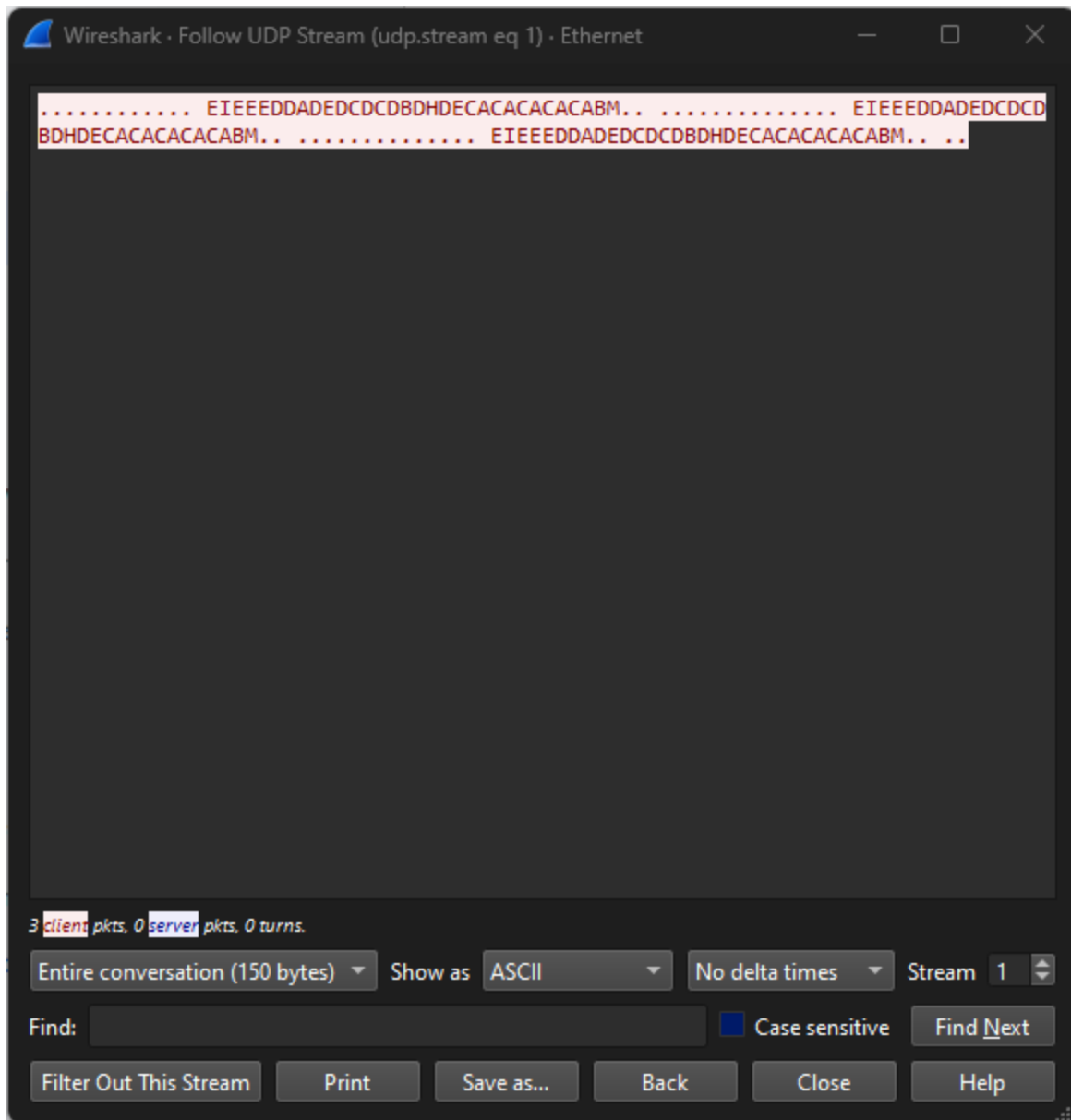
The image shows a Wireshark packet capture window titled "*Ethernet". The display filter is set to "udp.stream eq 1". The packet list shows three packets, all of which are NetBIOS Name Service (NBNS) queries from 172.16.53.154 to 172.16.53.255. The packet details pane shows the structure of the selected packet (Frame 2), including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and NetBIOS Name Service. The packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.056043	172.16.53.154	172.16.53.255	NBNS	92	Name query NB HDC0422174<10
25	0.819942	172.16.53.154	172.16.53.255	NBNS	92	Name query NB HDC0422174<10
60	1.579402	172.16.53.154	172.16.53.255	NBNS	92	Name query NB HDC0422174<10

Frame 2: Packet, 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface 0
Ethernet II, Src: EliteGroupCo_14:8a:3e (88:ae:dd:00:4e:3c), Dst: 01:00:5e:00:00:00
Internet Protocol Version 4, Src: 172.16.53.154, Dst: 172.16.53.255
User Datagram Protocol, Src Port: 137, Dst Port: 137
NetBIOS Name Service

0000 ff ff ff ff ff ff 88 ae dd 14 8a 3e 08 00 45
0010 00 4e 3c 2a 00 00 80 11 3a bb ac 10 35 9a ac
0020 35 ff 00 89 00 89 00 3a dd 9b f8 9f 01 10 00
0030 00 00 00 00 00 00 20 45 49 45 45 45 44 44 41
0040 45 44 43 44 43 44 42 44 48 44 45 43 41 43 41
0050 41 43 41 43 41 42 4d 00 00 20 00 01

wireshark_Ethernet1M32D3.pcapng Packets: 109 · Displayed: 3 (2.8%) · Dropped: 0 (0.0%) Profile: Default



RESULT

- Successfully captured and analyzed network traffic
- Identified specific protocols, conversations, and anomalies
- Gained insights into network performance, security posture, or protocol behavior